

Exercises on the Concurrency Abstraction and Interleavings
NOT ASSESSED

Instructions: This exercise sheet is not assessed - it is to help you understand the material, and to challenge what you think you know. Solutions for the following questions will be provided.

Collaboration: You are permitted to work with your classmates. However, you are likely to get more out of the following exercises if you attempt to solve them on your own.

Interleavings represent possible runtime executions of instructions from different threads. Given a set of threads running simultaneously, an *interleaving* is defined as a totally ordered sequence of atomic actions such that each action is performed by one thread, and actions within the same thread are executed in order.

1. Consider a concurrent system consisting of two threads *A* and *B*.
 - (a) Suppose that thread *A* includes two atomic actions *A1* and *A2*, while *B* includes only one atomic action *B1*. How many different interleavings exist?
 - (b) Suppose now that both *A* and *B* include two atomic actions each. List the possible interleavings we have this time.
2. Consider now a different concurrent system with three threads *A*, *B*, *C*.
 - (a) Suppose that each thread includes one atomic action – i.e., *A1*, *B1*, and *C1* respectively. List all the possible interleavings.
 - (b) Suppose that thread *A* includes two atomic actions *A1* and *A2*, *B* includes one atomic action *B1*, and *C* includes one atomic action *C1*. Enumerate the different interleavings that exist in this case.
3. Consider the case of a program that initializes an integer variable *x* to zero and then spawns two threads *T1* and *T2*.
 - (a) If *T1* executes the instructions $x=5$; $x=2*x$ and *T2* executes $x=x+2$, what values can *x* have after the two threads terminate?